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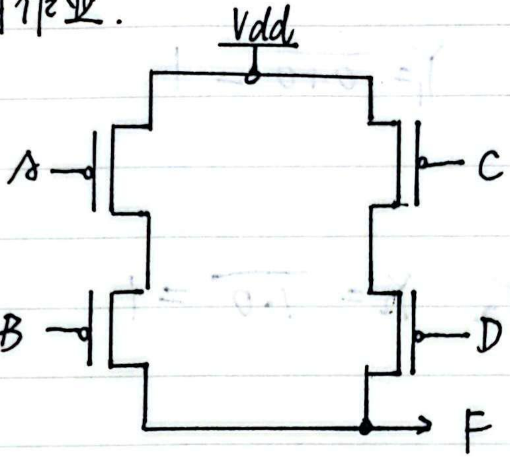
VLSI — 第二周作业.

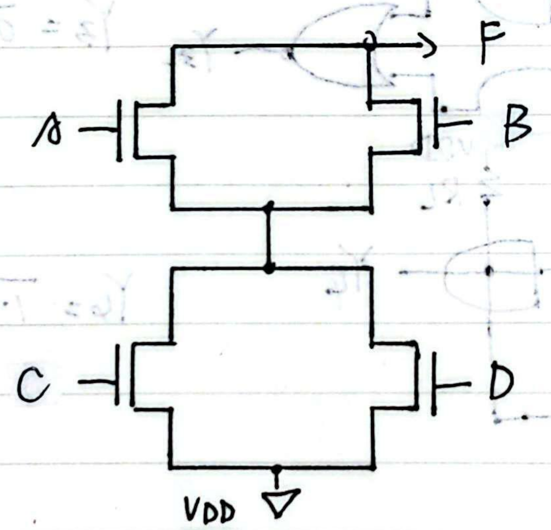
1. $F = A'B' + C'D'$

上拉网络:

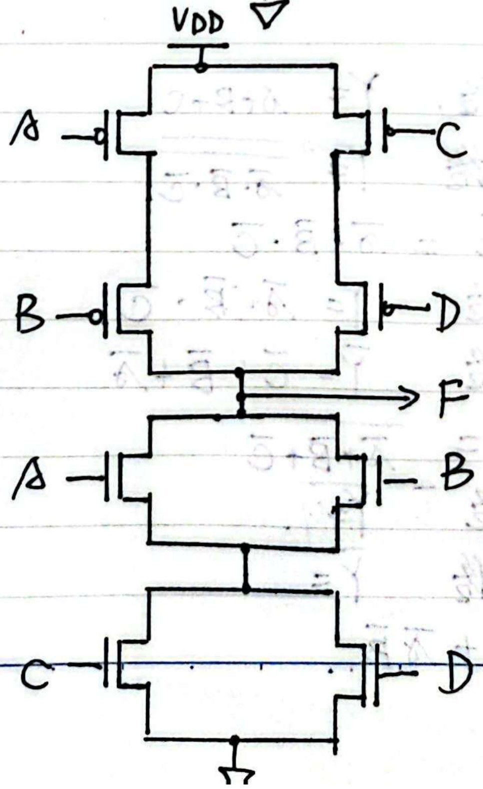


注: 有圈(VDD. F)

下拉网络 $\overline{F} = \overline{\overline{A}\,\overline{B} + \overline{C}\,\overline{D}} = (A+B)\cdot(C+D).$



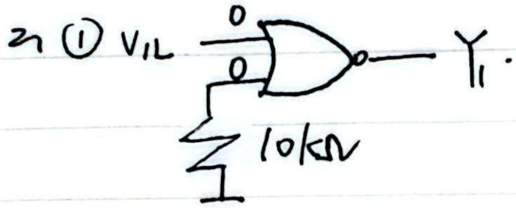
整体.



CMOS 不受影响.

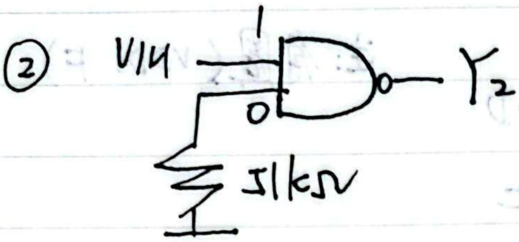
注意：CMOS 与TTL不同，~~CMOS大电阻接VDD=0~~ TTL大电阻接地=1

2. ①



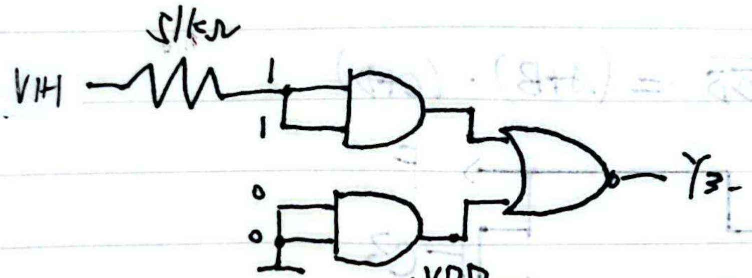
$Y_1 = \overline{0+0} = 1$

②



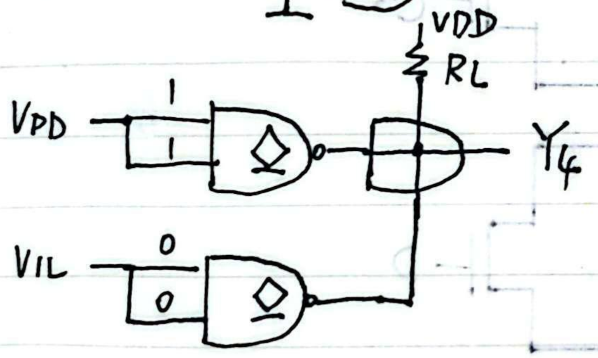
$Y_2 = \overline{1\cdot 0} = 1$

③



$Y_3 = \overline{\overline{1\cdot 1} + \overline{0\cdot 0}} = 0.$

④



$Y_4 = \overline{1\cdot 1} \cdot \overline{0\cdot 0} = 0$

3. (a). 经分析. 上拉网络. $Y = \overline{A+B+C}$

下拉网络 $Y = \overline{\overline{\bar{A}\cdot\bar{B}\cdot\bar{C}}}$

$\therefore Y = \overline{A+B+C} = \bar{A}\cdot\bar{B}\cdot\bar{C}$

(b) 经分析. 上拉网络 $Y = A\cdot B\cdot C$

下拉网络 $\bar{Y} = \bar{C} + \bar{B} + \bar{A}$

$\therefore Y = ABC = \overline{\bar{A}+\bar{B}+\bar{C}}$

(c). 经分析 $Y = \overline{AB}\cdot\overline{CD}\cdot\overline{INH}$.

(d) 经分析 $Y = AB + \bar{A}\bar{B}$